Theoretical Foundation of Hard Top-k Primal Routing

Polyhedral Exactness and Matroid Optimality in Sparse Mixture-of-Experts

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1 Introduction and Problem Formulation

A pervasive misconception in deep learning router design is that discrete Top-k selection operates as an ad-hoc, non-differentiable heuristic necessitating continuous relaxations such as softmax smoothing. Here, we formalize the token-to-expert dispatch mechanism through convex optimization and network flow theory, demonstrating that the hard Top-k operator over duality-adjusted affinities is the exact, closed-form solution to a linear program, incurring a zero integrality gap.

Let token-to-expert dispatch be formulated as a continuous optimization problem for a token i. Whether utilizing the duality-adjusted control signal from our Lagrangian framework or the heuristic selection bias from baseline models, any additive adjustment yields a net expert utility score Ψ_{ij}. Denote k the routing sparsity level (i.e., the number of selected experts per token) and E_r the number of conditionally routed experts. The continuous routing assignment vector $\mathbf{p}_i = [P_{i1}, \ldots, P_{iE_r}]^\top$ maximizes linear utility over the (E_r, k)-*hypersimplex polytope*, denoted as $\Delta(E_r, k)$ (defined as the convex hull of all binary vectors in $\{0,1\}^{E_r}$ whose entries sum to k):

$$\text{maximize} \quad \sum_{j=1}^{E_r} P_{ij}\Psi_{ij} \tag{1}$$

$$\text{subject to} \quad \sum_{j=1}^{E_r} P_{ij} = k, \tag{2}$$

$$0 \le P_{ij} \le 1, \quad \forall j \in \{1, \ldots, E_r\}. \tag{3}$$

This feasible region corresponds precisely to the uniform matroid base polytope of rank k [1].

2 Total Unimodularity and Matroid Optimality

By casting the constraint matrix of the hypersimplex in standard form, it can be shown to possess the total unimodularity property. Because the constraint matrix is totally unimodular and the boundary vector is integral, every extreme point (vertex) of the continuous polyhedron $\Delta(E_r, k)$ is strictly binary [1]:

$$\text{ext}(\Delta(E_r, k)) = \left\{ \mathbf{p} \in \{0,1\}^{E_r} : \sum_{j=1}^{E_r} p_j = k \right\}. \tag{4}$$

Consequently, the continuous relaxation has a zero integrality gap. Furthermore, maximizing a linear objective over a uniform matroid base polytope is solved globally and exactly by the

greedy algorithm. Sorting the affinities $\mathbf{\Psi}_i$ in descending order yields the exact binary vertex via the indicator condition:

$$P_{ij}^{*} = \begin{cases} 1, & j \in \mathcal{K}_i, \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

where $\mathcal{K}_i = \text{Top-k}_j(\Psi_{ij})$.

3 Remarks on Implementation and Router Design

This polyhedral exactness explains why auxiliary-loss-free routers utilizing hard Top-k achieve superior stability over continuous relaxations. Standard approximations (such as softmax over the expert set) smooth out discrete choices, introducing approximation slack and sensitivity into a problem that is analytically solvable as an exact linear program. Hard Top-k selection bypasses these approximations, directly executing the optimal extreme-point solution of the routing polytope.

From an implementation standpoint, computing $\mathcal{K}_i$ requires only an argsort or parallel reduction operation along the expert dimension, keeping computational complexity negligible while preserving theoretical optimality.

For a comprehensive treatment of the underlying control-theoretic framework, complete algebraic proofs, and system derivations of the Lagrangian Router, refer to the companion technical report [2]. Additionally, the complete PyTorch implementation, configuration scripts, and reproducible benchmarking suites are publicly archived in the open-source repository [3].

References

[1] Ravindra K. Ahuja, Thomas L. Magnanti, and James B. Orlin. *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall, Englewood Cliffs, NJ, 1993.

[2] Carl Johan Rehn. *Stochastic Lagrangian Control and Proportional Bias Routing for Mixture-of-Experts*, 2026. Available at https://github.com/lambda-z-lab/lz-moe-slc/blob/main/main.pdf. Zenodo DOI: https://doi.org/10.5281/zenodo.23214045.

[3] Carl Johan Rehn. Stochastic Lagrangian Control and Proportional Bias Routing for Mixture-of-Experts: Companion Source Code and Benchmarks. GitHub Repository, 2026. Available at https://github.com/lambda-z-lab/lz-moe-slc. Zenodo DOI: https://doi.org/10.5281/zenodo.23214045.